

AWS BASIC CONCEPTS

1. AWS Region

An AWS Region is a geographical area where AWS has multiple data centers. Each Region is independent and contains multiple Availability Zones. For example, ap-south-1 is the Mumbai AWS Region. You can choose a Region based on latency, cost, compliance, and service availability.

2. Availability Zone (AZ)

An Availability Zone is an isolated data-center location within an AWS Region. Each Region contains multiple Availability Zones, such as ap-south-1a, ap-south-1b, and ap-south-1c. Using multiple Availability Zones helps improve application availability and fault tolerance.

3. VPC (Virtual Private Cloud)

A VPC is a logically isolated virtual network in AWS where you can launch and manage AWS resources such as EC2 instances. It allows you to define your own IP address range, subnets, route tables, and network security. For example, you can create a VPC with the CIDR range 10.0.0.0/16.

4. CIDR (Classless Inter-Domain Routing)

CIDR is a method of defining an IP address range for networks and subnets. For example, 10.0.0.0/16 represents a large private IP address range that can be divided into smaller subnet ranges such as 10.0.1.0/24 and 10.0.2.0/24.

5. Subnet

A subnet is a smaller network created inside a VPC. Each subnet belongs to a single Availability Zone and has its own IP address range. Subnets are commonly classified as public or private based on their routing configuration.

6. Public Subnet

A public subnet is a subnet whose route table contains a route to an Internet Gateway. Resources such as public-facing EC2 instances and internet-facing load balancers can be placed in public subnets. An EC2 instance also requires a public IP address and appropriate security rules to communicate directly with the internet.

7. Private Subnet

A private subnet does not have a direct route to an Internet Gateway. It is commonly used to host backend applications, databases, and other resources that should not be directly accessible from the internet. Resources in private subnets can use a NAT Gateway for outbound internet access.

8. Route Table (RT)

A route table contains routing rules that determine where network traffic should be directed. Each route specifies a destination and a target. For example, the route 0.0.0.0/0 to an Internet Gateway allows traffic destined for the internet to be forwarded through that gateway.

9. Internet Gateway (IGW)

An Internet Gateway is a VPC component that enables communication between resources in a VPC and the internet. It is attached to a VPC and works with route tables to provide internet connectivity to resources in public subnets.

10. NAT Gateway

A NAT Gateway allows resources in private subnets to initiate outbound connections to the internet without allowing unsolicited inbound connections from the internet. It is commonly used when private EC2 instances need to download software updates or access external APIs.

11. Security Group

A Security Group acts as a virtual firewall for AWS resources such as EC2 instances. It controls inbound and outbound traffic using rules based on protocols, ports, and IP addresses. Security Groups are stateful, meaning response traffic for an allowed connection is automatically permitted.

12. Network ACL (NACL)

A Network Access Control List is a subnet-level firewall that controls inbound and outbound network traffic. It supports both allow and deny rules. Unlike Security Groups, NACLs are stateless, meaning inbound and outbound traffic must be allowed independently.

13. EC2 (Elastic Compute Cloud)

EC2 is an AWS compute service that provides virtual servers called instances. You can select the operating system, CPU, memory, storage, and networking configuration

according to your application requirements. EC2 is commonly used to host websites, APIs, databases, and applications.

14. AMI (Amazon Machine Image)

An AMI is a template used to launch EC2 instances. It contains the operating system and other configuration information required to create a virtual server. For example, an Amazon Linux AMI can be used to launch an EC2 instance running Amazon Linux.

15. EBS (Elastic Block Store)

EBS provides persistent block storage for EC2 instances. It works similarly to a virtual hard disk and is commonly used to store operating systems, application files, and databases. Data stored on an EBS volume can persist independently of the lifecycle of an EC2 instance.

16. Elastic IP

An Elastic IP is a static public IPv4 address that can be allocated to your AWS account and associated with supported AWS resources, such as EC2 instances. It can be reassigned to another resource when required.

17. S3 (Simple Storage Service)

S3 is an object storage service used to store and retrieve files and other data as objects inside buckets. It provides scalable and durable storage for images, videos, backups, logs, and application files. S3 also supports versioning, lifecycle policies, and different storage classes.

18. IAM (Identity and Access Management)

IAM is an AWS service used to manage authentication and authorization for AWS resources. It allows you to create users, groups, roles, and policies to control who can access AWS services and what actions they can perform.

19. IAM Role

An IAM Role is an AWS identity with specific permissions that can be assumed by users, applications, or AWS services. For example, an EC2 instance can assume an IAM role to access an S3 bucket without storing permanent AWS credentials.

20. IAM Policy

An IAM Policy is a JSON document that defines permissions for AWS resources. It specifies which actions are allowed or denied on particular resources. For example, a

policy can allow an IAM role to perform the `s3:GetObject` action on a specific S3 bucket.

21. Load Balancer

A Load Balancer distributes incoming network or application traffic across multiple targets, such as EC2 instances, containers, or IP addresses. It improves application availability, scalability, and fault tolerance.

22. Auto Scaling

Auto Scaling automatically adjusts the number of EC2 instances according to application demand or configured scaling policies. It can launch additional instances when traffic increases and terminate unnecessary instances when demand decreases.

23. CloudWatch

Amazon CloudWatch is a monitoring and observability service used to collect metrics, logs, and other operational information from AWS resources and applications. It can monitor CPU utilization, memory metrics collected through an agent, application logs, and other performance indicators. CloudWatch alarms can trigger notifications or automated actions.

24. ECR (Elastic Container Registry)

Amazon ECR is a managed container image registry used to store, manage, and retrieve Docker and other OCI-compatible container images. It is commonly used with ECS, EKS, and CI/CD pipelines.

25. ECS (Elastic Container Service)

Amazon ECS is a managed container orchestration service that allows you to deploy, run, and manage containerized applications. It supports running containers on EC2 instances or AWS Fargate without requiring Kubernetes.

26. EKS (Elastic Kubernetes Service)

Amazon EKS is a managed Kubernetes service that simplifies the deployment and management of Kubernetes clusters on AWS. AWS manages the Kubernetes control plane, while applications run on worker nodes or supported serverless compute infrastructure.

27. AWS Lambda

AWS Lambda is a serverless compute service that allows you to execute code without provisioning or managing servers. Lambda functions can run in response to events such

as API requests, S3 uploads, or EventBridge events.

28. CloudTrail

AWS CloudTrail records API calls and account activity across AWS services. It helps with security auditing, compliance, troubleshooting, and tracking actions performed by users, roles, and AWS services.

29. AWS CLI (Command Line Interface)

AWS CLI is a command-line tool that allows you to interact with AWS services directly from a terminal. It can be used to create, configure, monitor, and manage AWS resources without using the AWS Management Console.

30. Terraform

Terraform is an Infrastructure as Code tool used to provision and manage cloud infrastructure through configuration files. It allows you to define AWS resources such as VPCs, subnets, EC2 instances, and EKS clusters and manage them using commands such as `terraform init`, `terraform plan`, and `terraform apply`.